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FEDERAL SPECIFICATION

COATING, HEAT RESISTING

The General Services Administration has authorized the use of this federal specification by all federal agencies, some states and localities.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers heat resistant coatings that will withstand elevated service temperatures and exposure to corrosive environments.

1.2 Classification. Coatings shall be of the following types, classes, grades, compositions, and applications (see 6.2).

1.2.1 Types.

a. Type I – Type I coatings shall be liquid-applied coatings.

b. Type II – Type II coatings shall be powder-applied coatings.

1.2.2 Classes.

a. Class 1 – Class 1 coatings are intended for surfaces up to 1,200 degrees Fahrenheit (°F) (649 degrees Celsius [°C]).

b. Class 2 – Class 2 coatings are intended for surfaces up to 700 °F (371 °C).

c. Class 3 – Class 3 coatings are intended for surfaces up to 400 °F (204 °C).

d. Class 4 – Class 4 coatings shall be suitable as a thermal spray nonskid (TSN) topcoat.

1.2.3 Grades.

a. Grade A – Grade A coatings shall not require a heat cure to achieve full performance criteria.

b. Grade B – Grade B coatings shall require a heat cure to achieve full performance criteria.

1.2.4 Compositions.

a. Composition 1 – Composition 1 coatings shall be one-component systems.

b. Composition 2 – Composition 2 coatings shall be two-component systems.

1.2.5 Applications.

a. Application A – Application A coatings shall be suitable for exterior use.

a. Application B – Application B coatings shall be suitable for interior use.

Comments, suggestions, or questions on this document should be addressed to: Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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2. APPLICABLE DOCUMENTS

2.1 Government publications. The following documents, of the issues in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

FEDERAL STANDARDS

FED-STD-141 - Paint, Varnish, Lacquer and Related Materials: Methods of Inspection, Sampling and Testing

FED-STD-313 - Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities

(Copies of these document**/s** are available online at https://quicksearch.dla.mil/.)

FEDERAL REGULATIONS

40 CFR 60 - Determination of Volatile Matter Content, Water Content, Density, Volume Solids, and Weight Solids of Surface Coatings

40 CFR 63 - National Emission Standards for Hazardous Air Pollutants for Source Categories

40 CFR 82 - Protection of Stratospheric Ozone

40 CFR 261 - Identification and Listing of Hazardous Waste

(The code of Federal Regulations (CFR) are available for download at http://www.ecfr.gov. The Federal Register (FR) is available for download at the https://www.fedralregister.gov.)

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

T9070-AL-DPC-020/077-2 - NAVSEA Hazardous Material Avoidance Process

(Copies of this document are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/ by searching for the document number without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228 0669. This document is available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil.)

CENTER FOR DISEASE CONTROL (CDC), NATIONAL INSTITUTE FOR OCCUPATIONAL SAFETY AND HEALTH (NIOSH)

NIOSH Manual of Analytical Methods

(Copies of this document are available online at https://www.cdc.gov/niosh/.)

U.S. ENVIRONMENTAL PROTECTION AGENCY (EPA)

SW-846 - Hazardous Waste Test Methods

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI Z400.1/Z129.1 - Hazardous Workplace Chemicals – Hazard Evaluation and Safety Data Sheet and Precautionary Labeling Preparation

(Copies of this document are available online at http://webstore.ansi.org/.)

ASTM INTERNATIONAL

ASTM A1011/A1011M - Standard Specification for Steel, Sheet and Strip, Hot-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, and Ultra-High Strength

ASTM B117 - Standard Practice for Operating Salt Spray (Fog) Apparatus

ASTM D523 - Standard Test Method for Specular Gloss

ASTM D562 - Standard Test Method for Consistency of Paints Measuring Krebs Unit (KU) Viscosity Using a Stormer-Type Viscometer

ASTM D610 - Standard Practice for Evaluating Degree of Rusting on Painted Steel Surfaces

ASTM D1141 - Standard Practice for the Preparation of Substitute Ocean Water

ASTM D1640/D1640M Standard Test Methods for Drying, Curing, or Film Formation of Organic Coatings

ASTM D1654 - Standard Test Method for Evaluation of Painted or Coated Specimens Subjected to Corrosive Environments

ASTM D1729 - Standard Practice for Visual Appraisal of Colors and Color Differences of Diffusely-Illuminated Opaque Materials

ASTM D1849 - Standard Test Method for Package Stability of Paint

ASTM D2369 - Standard Test Method for Volatile Content of Coatings

ASTM D3359 - Standard Test Methods for Rating Adhesion by Tape Test

ASTM D3278 - Standard Test Methods for Flash Point by Small Scale Closed-Cup Tester

ASTM D4400 - Standard Test Method for Sag Resistance of Paints Using a Multinotch Applicator

ASTM D5895 - Standard Test Methods for Evaluating Drying or Curing During Film Formation of Organic Coatings Using Mechanical Recorders

ASTM D6450 - Standard Test Method for Flash Point by Continuously Closed Cup (CCCFP) Tester

ASTM D6677 - Standard Test Method for Evaluating Adhesion by Knife

ASTM F718 - Standard Specification for Shipbuilders and Marine Paints and Coatings Product/Procedure Data Sheet

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ASTM G154 - Standard Practice for Operating Fluorescent Ultraviolet (UV) Lamp Apparatus for Exposure of Nonmetallic Materials

(Copies of these documents are available online at https://www.astm.org.)

SAE INTERNATIONAL (SAE)

SAE AMS-STD-595 - Colors Used in Government Procurement

SAE AMS-STD-595/17178 - Silver/Aluminum

SAE AMS-STD-595/26008 - Deck Gray

SAE AMS-STD-595/26270 - Haze Gray

SAE AMS-STD-595/31136 - Red

SAE AMS-STD-595/33538 - Yellow

SAE AMS-STD-595/36076 - Dark Gray

SAE AMS-STD-595/37038 - Black

SAE AMS-STD-595/37875 - White

(Copies of these documents are available online at https://www.sae.org.)

SOCIETY FOR PROTECTIVE COATINGS (SSPC)

SSPC-SP 1 - Solvent Cleaning

(Copies of this document are available online at https://www.sspc.org.)

3. REQUIREMENTS

3.1 Qualification. The coatings furnished under this specification shall be products qualified for listing on the applicable qualified products list.

3.2 Composition (all types, all classes, all grades, all compositions, all applications). The coating materials supplied under this specification shall comply with all occupational health and safety regulations and shall meet the requirements of this specification.

3.2.1 Pigments (all types, all classes, all grades, all compositions, all applications). The selection of pigments shall be made so that the color of the dry coating visually matches the identified color standard in SAE AMS-STD-595 or the color identified by the manufacturer. The soluble and total metals content shall be not greater than the limits listed in table I.

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TABLE I. Metals content.

Metal and its compound in each coating dry film	Soluble metal, maximum (mg/L)	Total metal content, maximum (wt%)
Antimony	15.0	0.015
Arsenic	5.0	0.005
Barium (excluding barite)	100.0	0.10
Beryllium	0.75	0.0002
Cadmium	1.0	0.0005
Chromium (III)	560.0	0.56
Chromium (VI)	1.0	0.001
Cobalt 1/	50.0	0.005
Copper 2/	25.0	0.01
Fluoride salts	180.0	0.18
Lead	5.0	0.005
Mercury	0.2	0.0002
Molybdenum	350.0	0.035
Nickel	20.0	0.02
Selenium	1.0	0.002
Silver	5.0	0.001
Tantalum	100.0	0.100
Thallium	7.0	0.007
Tungsten	100.0	0.100
Vanadium	24.0	0.01
Zinc	250.0	0.25

NOTES:

1/ Total cobalt content may exceed 0.005 percent by weight (wt%) (up to 0.2 wt%) only if a cobalt dryer is used. However, soluble cobalt content shall not exceed the requirements of this table.

2/ Copper content does not apply to green and blue colors.

3.3 Identification (ID) characteristics (all types, all classes, all grades, all compositions, all applications). Values for ID characteristics shall be those established for the product at the time submitted for qualification testing (table II). The purpose of these values is to serve as a basis for determining that the coating being offered is essentially the same as that which was approved under qualification testing. Unless otherwise specified (see 6.2), ID characteristics for all types, classes, grades, compositions, and applications shall be conducted at 60 to 80 °F (16 to 27 °C) and below 85 percent relative humidity. The manufacturer shall provide ASTM methods, FED-STD-141 methods, other consensus standards, or a copy of unique test methodology or any other necessary information on methodology used to determine reported ID characteristics.

a. Color
b. Unit of issue
c. Percent nonvolatile vehicle
d. Percent volatiles (type I)

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e. Pot life for two-component systems
f. Maximum dry film thickness
g. Density for type I coatings (composition 1) or all liquid components (composition 2)
h. Chemical nature of each system component (e.g., epoxy, polysiloxane, epoxy powder, etc.)
i. Application method(s)
j. Mix ratios of each system component/Mixing instructions (composition 2)
k. Grade and heat cure instructions (see 1.2.3)
l. Shelf life
m. Maximum continuous service temperature
n. Maximum wet film thickness (WFT) per coat

TABLE II. ID characteristics and composition requirements.

Criteria	Classification	Requirements	Test method
Composition	All	3.2	---
Pigments	All	3.2.1	---
ID Characteristics	All	3.3	---
Vehicles	Type I	3.3.1.1	---
	Type II	3.3.1.2	---
Volatile organic compound (VOC) content	Type I	3.3.2.1	4.6.1
	Type II	3.3.2.2	4.6.1
Crystalline silica content	All	3.3.3.1	4.6.4
Metal content	All	3.3.3.2	4.6.3
Tantalum and tungsten content	All	3.3.3.3	4.6.5
Toxicity	All	3.3.4	4.6.2
Hazardous air pollutants (HAPS)	All	3.3.5	4.6.2
Prohibited materials	All	3.3.6	4.6.6
Safety data sheet	All	3.3.7	---
ASTM F718	All	3.3.7	---

3.3.1 Vehicles.

3.3.1.1 Type I (all classes, all grades, all compositions, all applications). Type I coating vehicles shall be chosen by the manufacturer so that the coating shall meet the requirements of the specification. The vehicle chosen shall be a silicone, modified silicone, or organosilicone resin or an epoxy-based material.

3.3.1.2 Type II (all classes, all grades, all compositions, all applications). Type II coating vehicles shall be a silicone or modified silicone powder dispersed in an epoxy powder with flow control agents as required to meet the requirements of this specification.

3.3.2 Volatile organic compound (VOC) content.

3.3.2.1 Type I (all classes [excluding class 4], all grades, all compositions, all applications). When tested in accordance with 4.6.1, the VOC content of the type I coatings shall be not greater than 3.5 pounds of solvent per gallon (lb/gal) (420 grams per liter [g/L]) of coating.

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3.3.2.2 Type II (all classes, all grades, all compositions, all applications). When tested in accordance with 4.6.1, the VOC content of the type II coatings shall be not greater than 0.63 lb/gal (75 g/L) of coating.

3.3.2.3 Class 4 (type I, class 4, all grades, all compositions, all applications). When tested in accordance with 4.6.1, the VOC content of the type I coatings shall be not greater than 2.8 lb/gal (340 g/L) of coating.

3.3.3 Hazardous pigments and additives.

3.3.3.1 Crystalline silica content (all types, all classes, all grades, all compositions, all applications). When tested in accordance with 4.6.4, crystalline silica shall be not greater than 1.0 wt% of the coating.

3.3.3.2 Metals content (all types, all classes, all grades, all compositions, all applications). When tested in accordance with 4.6.3, the metal content (soluble and total) of the coating shall be not greater than the values listed in table I. If the test for total metal content results in a value less than the soluble metal limit, the soluble metal test need not be conducted.

3.3.3.3 Tantalum and tungsten content (all types, all classes, all grades, all compositions, all applications). When tested in accordance with 4.6.5, the tantalum or tungsten content shall be not greater than the values listed in table I.

3.3.4 Toxicity (all types, all classes, all grades, all compositions, all applications). When evaluated in accordance with 4.6.2, the coating shall pose no serious or high risk to the health of personnel or the environment when used for its intended purpose (see 4.6.2 and 6.4).

3.3.5 Hazardous air pollutants (HAPS) (all types, all classes, all grades, all compositions, all applications). Unless otherwise specified (see 6.2), the content of the HAPS solvents in the coating shall be not greater than the wt% values listed in table III. Unless otherwise specified within these limitations and the requirement that the finished coating meet all requirements herein, solvent selection is the responsibility of the manufacturer. The definition of HAPS materials is in accordance with 40 CFR 63.

TABLE III. Hazardous solvents.

Hazardous solvent in mixed coating or its components	**Maximum (wt%)**
Benzene	0.05
Chlorinated solvent(s), total	0.05
Ethyl benzene	0.05
Methyl-, ethyl-, and butyl- mono-ethers of ethylene glycol or the acetates thereof, total (also known as methyl-, ethyl-, and butyl- cellosolve acetates)	0.05
Methyl ethyl ketone	0.05
Methyl isobutyl ketone	0.05
Solvents containing fluorine in accordance with 40 CFR 82	0.01
Toluene	0.05
Xylene (all forms), total	0.1

3.3.6 Prohibited materials (all types, all classes, all grades, all compositions, and all applications). When tested in accordance with 4.6.6, the coating system shall not contain any chemicals categorized as “prohibited” in accordance with T9070-AL-DPC-020/077-2.

3.3.7 Safety data sheet and ASTM F718 (all types, all classes, all grades, all compositions, all applications). The contractor shall provide to the qualifying activity a completed safety data sheet (SDS) in accordance with ANSI Z129.1 or FED-STD-313. An ASTM F718 for the coating shall be prepared by the manufacturer for NAVSEA review.

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3.4 Quantitative requirements. When tested in accordance with 4.7, the coating shall conform to the quantitative requirements specified in 3.4.1 through 3.4.9 and table IV.

TABLE IV. Quantitative requirements.

Characteristic	Classification	Standard	Requirements	Test method	Conformance requirements
Color	Application A Excluding Class 4	ASTM D1729	3.4.1.1	4.7.1.1	Yes
	Application B	ASTM D1729	3.4.1.2	4.7.1.2	Yes
	Class 4	ASTM D1729	3.4.1.3	4.7.1.3	Yes
Gloss	Application A Excluding Class 4	ASTM D523	3.4.2.1	4.7.1.2	Yes
	Class 4	ASTM D523	3.4.2.2	4.7.2.2	Yes
Total solids	Type I	ASTM D2369	3.4.3	4.7.3	Yes
Flash point	Type I Application A	ASTM D3878/ ASTM D6450	3.4.4	4.7.4	Yes
	Type I Application B	ASTM D3278/ ASTM D6450	3.4.4.2	4.7.4	Yes
Drying or cure time	Type I	ASTM D5895	3.4.5	4.7.5.1	Yes
	Type I, Grade B	FED-STD-141, method 6304.2	3.4.5	4.7.5.2	Yes
	Type II	FED-STD-141, method 6304.2	3.4.5	4.7.5.3	Yes
Adhesion	Application A Excluding Class 4	ASTM D6677	3.4.6.1	4.7.6.1	No
	Application B	ASTM D6677	3.4.6.2	4.7.6.2	No
	Class 4	ASTM D3359, method B	3.4.6.3	4.7.6.3	No
Corrosion resistance	Application A Excluding Class 4	ASTM D1654, method A	3.4.7	4.7.7	No
		ASTM D610			No
Viscosity	Type I Excluding Class 4	ASTM D562	3.4.8	4.7.8	Yes
	Type I Class 4	ASTM D562	3.4.8.1	4.7.8	Yes
Coefficient of friction	Class 4	N/A	3.4.9	4.7.9	No

3.4.1 Color (all types, all classes, all grades, all compositions, all applications). When tested in accordance with 4.7.1, the color of dried coatings shall visually match the following conditions (see 6.2).

3.4.1.1 Application A (all types, all classes [excluding class 4], all grades, all compositions). When tested in accordance with 4.7.1.1, the color of the dry application A coating shall visually match one of the SAE AMS-STD-595 colors: 17178 Silver/Aluminum, 26008 Deck Gray, 26270 Haze Gray, or to the color identified or as defined by the manufacturer.

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3.4.1.2 Application B (all types, all classes, all grades, all compositions). When tested in accordance with 4.7.1.2, the color of the dry application B coating shall visually match the specified SAE AMS-STD-595 color or match the color defined by the manufacturer.

3.4.1.3 Class 4 (application A, all types, all grades, all compositions). When tested in accordance with 4.7.1.3, the color of the dry class 4 coating shall visually match one of the SAE AMS-STD-595 colors: 36076 Dark Gray, 33538 Yellow, 37875 White, 37038 Black, or 31136 Red.

3.4.2 Gloss (application A, all types, all classes, all grades, all compositions).

3.4.2.1 Application A (all types, all classes [excluding class 4], all grades, all compositions). When tested in accordance with 4.7.2.1, the as-applied, initial 60-degree specular gloss of application A coatings shall be not greater than 60 gloss units.

3.4.2.2 Class 4 (application A, all types, all grades, all compositions). When tested in accordance with 4.7.2.2, the as-applied, initial 60-degree specular gloss of application A coatings shall be not greater than 12 gloss units.

3.4.3 Total solids (type I, all classes, all grades, all compositions, all applications). When tested in accordance with 4.7.3, the total solids content shall be greater than 40 percent by weight of coating.

3.4.4 Flash point.

3.4.4.1 Flash point (type I, application A, all classes, all grades, all compositions). When tested in accordance with 4.7.4, the flash point of type I coatings shall be not less than 80 °F (27 °C).

3.4.4.2 Flash point (type I, application B, all classes, all grades, all compositions). When tested in accordance with 4.7.4, the flash point of type I coatings shall be not less than 100 °F (38 °C).

3.4.5 Drying or cure time (all types, all classes, all grades, all compositions, all applications). When tested in accordance with 4.7.5, the drying time of coatings shall be not greater than the times specified in table V.

TABLE V. Dry or cure times.

Type	Dry or cure conditions	Dry time
Type I	Dry-to-touch	Not greater than 4 hours
	Dry-through	Not greater than 16 hours
Type I, Grade B	Heat treatment	Not greater than 2 hours
Type II	Heat treatment	Not greater than 20 minutes

3.4.6 Adhesion.

3.4.6.1 Application A (all types, all classes [excluding class 4], all grades, all compositions). When tested in accordance with 4.7.6.1, the coatings shall be applied over the substrates with no visible film irregularities. The coating shall not wrinkle or lift from the test panel and shall dry to a smooth, uniform finish. The coating adhesion shall achieve a rating of 6.

3.4.6.2 Application B (all types, all classes [excluding class 4], all grades, all compositions). When tested in accordance with 4.7.6.2, the coatings shall be applied over the substrates with no visible film irregularities. The coating shall not wrinkle or lift from the test panel and shall dry to a smooth, uniform finish. The coating adhesion shall achieve a rating of 6.

3.4.6.3 Class 4 (all types, all grades, all compositions, all applications). When tested in accordance with 4.7.6.3, the coating shall have a minimum rating of 3B.

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3.4.7 Corrosion resistance (application A, all types, all classes [excluding class 4], all grades, all compositions). When tested in accordance with 4.7.7, the application A coatings shall have a pull-back from the scribe no greater than 0.5 inch (13 millimeters). The coated test panel shall show no corrosion, creepage, or undercutting beyond 0.5 inch (13 millimeters) from the scribe mark. The scribe line areas shall have ratings in accordance with ASTM D1654, method A of no less than 6. At all other points of the test panel, there shall be no more than a trace amount of rusting in accordance with ASTM D610, and no more than five scattered blisters, none of which is larger than 40 mils (1 millimeter) in diameter.

3.4.8 Viscosity (type I, all classes [excluding class 4], all grades, all compositions, all applications). When tested in accordance with 4.7.8, the viscosity of freshly opened and mixed type I coating shall be not less than 55 equivalent Krebs units and not greater than 120 Krebs units.

3.4.8.1 Viscosity (type I, class 4, all grades, all compositions, all applications). When tested in accordance with 4.7.8, the viscosity of a freshly opened and mixed type I, class 4 coating shall be not less than 55 equivalent Krebs units and not greater than 75 Krebs units.

3.4.9 Coefficient of friction (COF) (type I, class 4, all grades, all compositions, all applications). When tested in accordance with 4.7.9, the dynamic COF of the dark gray coating, when tested using a rotating arm COF meter, shall be no less than 1.0 on dry test panels and no less than 0.6 on wet panels.

3.5 Qualitative requirements. When tested in accordance with 4.8, the coating shall conform to the qualitative requirements specified in 3.5.1 through 3.5.8 and table VI.

TABLE VI. Qualitative requirements.

Characteristic	Classification	Standard	Requirements	Test method	Conformance requirements
Heat resistance	All	N/A	3.5.1	4.8.1	Yes
Salt spray resistance	Application A	ASTM B117	3.5.2	4.8.2	Yes
Storage stability	Type I	ASTM D1849 FED-STD-141, method 3011.3	3.5.3.1.1	4.8.3.1.1	No
	Type II	ASTM D1849 FED-STD-141, method 3011.3	3.5.3.1.2	4.8.3.1.2	No
	Type I, Composition 1	FED-STD-141, method 3022.2	3.5.3.2	4.8.3.2	No
Sag resistance	Type I Excluding Class 4	ASTM D4400	3.5.4	4.8.4	No
Fluid resistance	All	N/A	3.5.5	4.8.5	No
Recoating	Type II	ASTM B117	3.5.6	4.8.6	No
Brushing properties	Type I	FED-STD-141, method 4321.3	3.5.7	4.8.7	No
Spray properties	Type I	FED-STD-141, method 4331.2 method 2131.2	3.5.8.1	4.8.8.1	No
	Type II	---	3.5.8.2	4.8.8.2	No
Color retention	Application A	ASTM D1729	3.5.9	4.8.9	No

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3.5.1 Heat resistance (all types, all classes, all grades, all compositions, all applications). When tested in accordance with 4.8.1, the coating shall show no cracking, flaking, blistering, peeling, or chipping after being exposed to the required heat cycle for the identified class on any of the test panels. The coating may exhibit a reduction in gloss and the resulting color of the coating shall be visually similar to the color identified in 3.3a.

3.5.2 Salt spray resistance (application A, all types, all classes [excluding class 4], all grades, all compositions). When tested in accordance with 4.8.2, the coating shall show no cracking, flaking, or chipping after being exposed to salt spray and heat cycles. Examined immediately after removal from the test, the coating shall show no more than a trace amount of rusting and no more than five scattered blisters, none of which is larger than 40 mils (1 millimeter) in diameter. Upon removal of the coating from the test panels, the surface of the steel shall exhibit a Rust Grade of 10 (i.e., no rust), 9-S, 9-G, or 9-P in accordance with ASTM D610.

3.5.3 Storage stability.

3.5.3.1 Full container (all types, all classes, all grades, all compositions, all applications). When tested in accordance with 4.8.3.1, the coating shall exhibit a level 6 for all qualities in accordance with ASTM D1849.

3.5.3.1.1 Type I (all classes, all grades, all compositions, all applications). When tested in accordance with 4.8.3.1.1, a full, unopened coating container shall show no “gas” pressure or bulging, and the paint shall be free from grit, seeds, skins, lumps, abnormal thickening, or livering and shall show no more pigment settling or caking than can be readily reincorporated to a smooth uniform state by manual mixing.

3.5.3.1.2 Type II (all classes, all grades, all compositions, all applications). When tested in accordance with 4.8.3.1.2, the type II coating shall be removed from the container and shall show no solidification and no hard dry caking or tough gummy sediment.

3.5.3.2 Partially full container (type I, composition 1, all classes, all grades, all applications). When tested in accordance with 4.8.3.2, neither sample shall show “gas” pressure build-up or bulging of the container and the paint in both containers shall be free from grit, seeds, skins, lumps, abnormal thickening, and livering. The paint in both containers shall show no more pigment settling or caking than can be readily reincorporated to a smooth uniform state by manual mixing.

3.5.4 Sag resistance (type I, all classes [excluding class 4], all grades, all compositions, all applications). When tested in accordance with 4.8.4, the coating shall not sag when applied at twice the manufacturer’s maximum WFT.

3.5.5 Fluid resistance (all types, all classes, all grades, all compositions, all applications). When tested in accordance with 4.8.5, the applied coating shall show no wrinkling or blistering immediately after removal from the following liquids (see 3.5.5.1 and 3.5.5.2).

3.5.5.1 Hydrocarbon. When tested in accordance with 4.8.5.1, a coated test panel shall show no wrinkling or blistering immediately after removal from the JP-8 immersion. After 24 hours of air drying at standard laboratory conditions (see 4.5.1), the immersed test panel shall be almost indistinguishable with regard to hardness and adhesion from a test panel that was prepared at the same time but not immersed.

3.5.5.2 Potable water. When tested in accordance with 4.8.5.2, a coated test panel shall show no wrinkling or blistering immediately after removal from potable water immersion. The coating shall show no wrinkling or blistering immediately after removal and 2 hours after removal from the potable water. After 24 hours of air drying at standard laboratory conditions (see 4.5.1), the coating shall show no whitening or dulling in comparison to the non-immersed coated test panel.

3.5.6 Recoating (type II, all classes, all grades, all compositions, all applications). When tested in accordance with 4.8.6, the fully cured type I coating shall show no blistering, wrinkling, or lifting when used to recoat the type II coating.

3.5.7 Brushing properties (type I, all classes [excluding class 4], all grades, all compositions, all applications). When tested in accordance with 4.8.7, the coating shall brush easily with good flow and leveling properties and shall dry to a smooth, uniform film free of seeds, runs, sags, or streaks.

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3.5.8 Spray properties.

3.5.8.1 Type I (all classes [excluding class 4], all grades, all compositions, all applications). When tested in accordance with 4.8.8.1, the coating shall spray satisfactorily in all respects and shall show no running, sagging, or streaking. The dried film shall show no dusting or mottling and shall present a smooth, uniform finish free of seeds.

3.5.8.2 Type II (all classes, all grades, all compositions, all applications). When tested in accordance with 4.8.8.2, the dry film thickness shall be 3 mils or the minimum coating thickness cited in the manufacturer's commercial product data sheet for powder coatings to meet the requirements of this specification. The cured film shall be uniform, smooth, even, and free of runs, sags, blow holes, pores, and streaks.

3.5.9 Color retention (application A, all types, all classes, all grades, all compositions). When tested in accordance with 4.8.9, the coating shall show no cracking, flaking, blistering, or peeling on the test panels. The coating may exhibit a reduction in gloss and the resulting color of the coating shall be visually similar to the color identified in 3.3a.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of examinations and tests.

a. Qualification inspection (4.3).

b. Quality conformance inspection (4.4).

4.3 Qualification inspection. Qualification inspection shall be conducted at a laboratory acceptable to the qualifying activity. Qualification inspection shall consist of all of the tests required by this specification and specified in 4.3.1, 4.3.2, 4.6, and tables IV and VI.

4.3.1 General inspection. The coating system's ID characteristics shall be reviewed to ensure conformance to the requirements specified in 3.3.

4.3.2 Formulation inspection. The coating system's composition (3.2), pigmentation (see 3.2.1), and vehicle (see 3.3.1), shall be verified as required by the qualifying activity.

4.3.3 Change approval. Changes in construction, formulation, materials, design, or manufacturing process of qualified products may require requalification. Unless otherwise specified by the qualifying activity, changes require written approval by the qualifying activity (see 6.5). Incorporation of any changes that have not been so approved shall require requalification of the item in question.

4.4 Quality conformance inspection. Unless otherwise specified (see 6.2), quality conformance inspection testing shall consist of the tests specified in tables IV and VI. At a minimum, quality conformance inspections shall be performed on the first lot and every 5,000 gal (19,000 L) thereafter.

4.4.1 Lot. A lot shall consist of an individual system component of the same type, class, grade, composition, and application from a single, uniform batch or a uniform blend of batches offered for delivery at one time.

4.5 Standard conditions and procedures.

4.5.1 Standard laboratory conditions. Unless otherwise specified herein, all preparation, application, and testing shall be conducted under standard laboratory conditions. Standard laboratory conditions are to be maintained at 60 to 80 °F (16 to 27 °C) and 40 to 60 percent relative humidity.

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4.5.2 Test panel preparation. Unless otherwise specified herein, test panels shall be 4 by 6 by 0.125 inches (nominal) (102 by 152 by 3 millimeters), rust free, cold-rolled steel plates from steel in accordance with ASTM A1011/1011M. The test panels shall be hand sanded or orbital sanded with 80 to 120 aluminum oxide grit paper to create a uniformly distributed random pattern of scratches on the surface. Prior to coating, the test panels shall be solvent cleaned and be in accordance with cleanliness requirements of SSPC-SP 1.

4.5.2.1 Coating procedure – type I. Unless otherwise specified herein, coatings shall be mixed and applied in accordance with the manufacturer's ASTM F718. All type I films shall be applied at the manufacturer's maximum WFT. Type I coatings shall be allowed to acclimate to standard laboratory conditions (see 4.5.1) for a minimum of 24 hours prior to use. Composition 1 and 2 coatings shall be shaken for a minimum of 3 minutes. Composition 2 coatings shall be mixed in accordance with the manufacturer's commercial product data sheet and mixed with a pneumatic or powered mixer. The type I, class 1, 2, and 3 coatings shall be spray-applied on all panel sides and edges per the manufacturer's ASTM F718. Type I, class 4 coatings shall be roller-applied with a ⅜-inch nap or split foam roller to the panels at the minimum thickness required to cover the substrate and provide a uniform appearance free of bare areas, blow holes, and pores. The coated panels shall be allowed to dry at standard laboratory conditions for no more than 24 hours between coats in accordance with 4.5.1. Unless otherwise specified (see 6.2), panels with grade B coatings shall be heat cured within 24 to 48 hours of coating application in accordance with manufacturer's ASTM F718 product data sheet. Unless otherwise specified (see 6.2), panels with grade A coatings shall not be heat cured.

4.5.2.2 Coating procedure – type II. Unless otherwise specified herein, type II powder coatings shall be applied in accordance with the manufacturer's commercial product data sheet. Type II coatings shall be allowed to acclimate to standard laboratory conditions (see 4.5.1). Type II coating test films shall be applied at a minimum film thickness of 3 mils or applied at the minimum coating thickness cited in the manufacturer's commercial product data sheet for powder coatings to meet the requirements of this specification. The coating shall be applied on all sides and edges of the panel in accordance with the electrostatic powder coating procedure cited in the manufacturer's commercial product data sheet.

4.5.3 Coefficient of friction test panels. Two square aluminum test panels that are each 18 inches by 18 inches by 0.25 inch (nominal) shall be hand sanded or orbital sanded with 80 to 120 aluminum oxide grit paper to create a uniformly distributed random pattern of scratches on the surface. Prior to coating, the test panels shall be solvent cleaned and be in accordance with cleanliness requirements of SSPC-SP 1. The panels shall be coated in accordance with 4.5.2.1.g. The panels shall be allowed to rest as necessary to reach "cure to service" time identified by the manufacturer.

4.6 Test procedures. Tests shall be conducted in accordance with the current version of FED-STD-141, applicable ASTM methods, or as specified herein. Products that fail to pass any test or exhibit non-compliance with the requirements of Section 3 shall not be qualified.

4.6.1 VOC content (all types, all classes, all grades, all compositions, all applications). VOC content for all types, classes, grades, compositions, and applications shall be determined in accordance with 40 CFR 60, Appendix A, method 24. Type I shall match the requirements for 3.3.2.1. Type II shall match the requirements for 3.3.2.2.

4.6.2 Toxicity (all types, all classes, all grades, all compositions, all applications). A Health Hazard Assessment (HHA) will be conducted to ensure conformance to 3.3.4 as required by the qualifying activity. The Navy and Marine Corps Public Health Center (NMCPHC) will evaluate the coating system using data provided by the manufacturer/distributor to the NMCPHC (see 3.3.4 and 6.4).

4.6.3 Hazardous pigments and additives (all types, all classes, all grades, all compositions, all applications). Soluble and total metal content shall be determined on a dry coating film made from each individual coating of the coating system in accordance with 40 CFR 261.24(a) and an appropriate test method from SW-846. Test results shall be in accordance with the requirements of 3.3.3. Calculation of individual hazardous metal content can be based on either the manufacturer's testing of batches in accordance with 40 CFR 261.24(a), or the supplier's data for raw materials used in the product. Soluble metal content shall be reported as milligrams per liter (mg/L).

4.6.4 Crystalline silica content (all types, all classes, all grades, all compositions, all applications). Crystalline silica shall be analyzed in accordance with the NIOSH Manual of Analytical Methods, method 7500 with the coating sample treated as a bulk settled dust. Test results shall meet the requirements of 3.3.3.1.

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4.6.5 Tantalum and tungsten content (all types, all classes, all grades, all compositions, all applications). The tantalum and tungsten content of the cured coating shall be determined using any appropriate spectroscopy test method. The tests shall be conducted in accordance with the instrument manufacturer's commercial product data sheet. Data supporting the test method choice and analytical accuracy shall be established. The test results for tantalum or tungsten shall meet the requirements of 3.3.3.3.

4.6.6 Prohibited materials. Prohibited materials shall be verified for conformance to 3.3.6 as required by the qualifying activity.

4.7 Quantitative testing.

4.7.1 Color (all types, all classes, all grades, all compositions, all applications).

4.7.1.1 Application A (excluding class 4). The test panels shall be prepared in accordance with 4.5.2 and coated in accordance with 4.5.2.1 or 4.5.2.2. The color of the dry coated test panel shall be compared to the identified SAE AMS-STD-595 color chip or card for 17178 Silver/Aluminum, 26270 Haze Gray, 26008 Deck Gray, or to the color identified by the manufacturer. Assessment of color matching shall be made in accordance with ASTM D1729. The results shall meet the requirements of 3.4.1.1.

4.7.1.2 Application B. The test panels shall be prepared in accordance with 4.5.2 and coated in accordance with 4.5.2.1 or 4.5.2.2. The color of the dry coated test panel shall be compared to the identified SAE AMS-STD-595 color chip or card or to the color as defined by the manufacturer. Assessment of color matching shall be made in accordance with ASTM D1729. The results shall meet the requirements of 3.4.1.2.

4.7.1.3 Class 4. The test panels shall be prepared in accordance with 4.5.2 and coated in accordance with 4.5.2.1. The color of the dry coated test panel shall be compared to the identified SAE AMS-STD-595 color chip or card for 36076 Dark Gray, 33538 Yellow, 37875 White, 37038 Black, or 31136 Red. Assessment of color matching shall be made in accordance with ASTM D1729. The results shall meet the requirements of 3.4.1.3.

4.7.2 Gloss (type I, application A, all classes, all grades, all compositions).

4.7.2.1 Application A (type I, all classes [excluding class 4], all grades, all compositions). The thoroughly mixed coating shall be applied to a standard black and white Leneta chart at the maximum WFT cited on the manufacturer's ASTM F718. The coated Leneta chart shall be allowed to dry for 1 day. The 60-degree gloss of the coated Leneta chart shall be measured in accordance with ASTM D523. The results shall meet the requirements of 3.4.2.1.

4.7.2.2 Class 4. The thoroughly mixed coating shall be applied to a standard black and white Leneta chart at the manufacturer's maximum WFT. The coated Leneta chart shall be allowed to dry for 1 day. The 60-degree gloss of the coated Leneta chart shall be measured in accordance with ASTM D523. The results shall meet the requirements of 3.4.2.2.

4.7.3 Total solids (all types, all classes, all grades, all compositions, all applications). The testing of the coatings shall be in accordance with ASTM D2369. The results shall meet the requirements of 3.4.3.

4.7.4 Flash point (all types, all classes, all grades, all compositions, all applications). The flash point of the coating (composition 1) or any component (composition 2) shall be determined in accordance with ASTM D3278, ASTM D6450, or through a method approved by the qualifying activity. The results shall meet the requirements of 3.4.4.

4.7.5 Drying or cure time (all types, all classes, all grades, all compositions, all applications).

4.7.5.1 Type I, grade A. The drying time shall be determined using a mechanical recorder in accordance with ASTM D5895. The coating shall be applied to a glass substrate at the maximum WFT cited on the manufacturer's ASTM F718 data sheet. All preparation, application, and testing shall be conducted under standard laboratory conditions (see 4.5.1). The dry-to-touch and dry-through times for the type I coating shall be determined in accordance with ASTM D5895. The test results shall meet the requirements of 3.4.5.

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4.7.5.2 Type I, grade B. The test panels shall be prepared in accordance with 4.5.2 under standard laboratory conditions (see 4.5.1). The coating shall be spray applied per the manufacturer's commercial product data sheet or roller applied with a ⅜-inch nap roller at the maximum WFT cited on the manufacturer's ASTM F718 data sheet. The coated test panel shall be heat cured in accordance with the manufacturer's ASTM F718 product data sheet. The film shall be considered to have reached full hardness when the coating meets the dry-through (or dry-to-handle) time requirement of ASTM D1640/D1640M. The test results shall meet the requirements of 3.4.5.

4.7.5.3 Type II. The test panels shall be prepared in accordance with 4.5.2 under standard laboratory conditions (see 4.5.1) and coated in accordance with 4.5.2.2. The film shall be considered to have reached full hardness when the coating meets the dry-through (or dry-to-handle) time requirement of ASTM D1640/D1640M. The test results shall meet the requirements of 3.4.5.

4.7.6 Adhesion.

4.7.6.1 Application A (all types, all classes, all grades, all compositions). Three test panels shall be prepared in accordance with 4.5.2 and coated in accordance with 4.5.2.1 or 4.5.2.2. One test panel shall be assessed in accordance with ASTM D6677 and compared to the requirements of 3.4.6.1. The remaining two test panels shall be exposed to 100 hours of salt spray in accordance with ASTM B117. The test panels shall be allowed to dry at standard laboratory conditions (see 4.5.1). The test panels shall be placed in an oven on a suitable rack or holder so that no parts of the test panels are in direct contact with the bottom or sides of the oven and subject to the required temperature for the coating's respective class as shown in table VII for 8 hours. The test panels shall be allowed to cool to standard laboratory conditions (see 4.5.1). Adhesion testing in accordance with ASTM D6677 shall be performed on the second test panel and examined for conformance to the requirements of 3.4.6.1. The remaining test panel shall be exposed to an additional four cycles of ASTM B117 salt spray and heat. The test results shall meet the requirements of 3.4.6.1.

TABLE VII. Heat resistance testing.

Coating class	Heat temperature
Class 1	1,200 °F (649 °C)
Class 2	700 °F (371 °C)
Class 3	400 °F (204 °C)
Class 4	350 °F (177 °C)

4.7.6.2 Application B (all types, all classes [excluding class 4], all grades, all compositions). Three test panels shall be prepared in accordance with 4.5.2 and coated in accordance with 4.5.2.1 or 4.5.2.2. One test panel shall be assessed in accordance with ASTM D6677 and compared to the requirements of 3.4.6.2. The test panels shall be placed in an oven on a suitable rack or holder so that no parts of the test panels are in direct contact with the bottom or sides of the oven and subject to the required temperature for the coating's respective class as shown in table VII for 8 hours. The test panels shall be allowed to cool to standard laboratory conditions (see 4.5.1). Adhesion testing in accordance with ASTM D6677 shall be performed on the second test panel and examined for conformance to the requirements of 3.4.6.2. The third test panel shall be exposed to an additional four cycles of heating and cooling to standard laboratory conditions (see 4.5.1). The test results shall meet the requirements of 3.4.6.2.

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4.7.6.3 Class 4 (all types, class 4, all grades, all compositions). Three test panels shall be prepared in accordance with 4.5.2 and coated in accordance with 4.5.2.1. One test panel shall be assessed in accordance with ASTM D3359, method B and compared to the requirements of 3.4.6.3. The remaining two test panels shall be placed in an oven on a suitable rack or holder so that no parts of the test panels are in direct contact with the bottom or sides of the oven and subject to the required temperature for the coating's respective class as shown in table VII for 8 hours. The test panels shall be allowed to cool to standard laboratory conditions (see 4.5.1). Adhesion testing in accordance with ASTM D3359, method B shall be performed on the second test panel and examined for conformance to the requirements of 3.4.6.3. The remaining test panel shall be exposed to an additional four cycles of heating and cooling to standard laboratory conditions (see 4.5.1). Adhesion testing in accordance with ASTM D3359, method B shall be performed on the third test panel and examined for compliance with 3.4.6. The test results shall meet the requirements of 3.4.6.3.

4.7.7 Corrosion resistance (application A, all types, all classes [excluding class 4], all grades, all compositions). Three test panels shall be prepared in accordance with 4.5.2. The test panels shall be coated on all sides and edges in accordance with 4.5.2.1 or 4.5.2.2. Grade B coatings shall be given a heat cure in accordance with their manufacturer's commercial product data sheet. Test panels shall have an "X" scribe cut in the coating down to the base metal. The test panels shall be exposed to 100 hours of salt spray in accordance with ASTM B117. The test panels shall be allowed to dry overnight. The test panels shall be placed in an oven on a suitable rack or holder so that no parts of the test panels are in direct contact with the bottom or sides of the oven and subject to the required temperature for the coating's respective class as shown in table VII for 8 hours. The test panels shall be allowed to cool to standard laboratory conditions (see 4.5.1). The test panels shall be exposed to an additional four cycles of heat and salt spray. The test results shall meet the requirements of 3.4.7.

4.7.8 Viscosity (type I, all classes, all grades, all compositions, all applications). The viscosity of a freshly opened and mixed coating shall be measured at standard laboratory conditions (see 4.5.1) in accordance with ASTM D562. The test results shall meet the requirements of 3.4.8.

4.7.9 Coefficient of friction (type I, class 4, all grades, all compositions, all applications). The rotating arm COF meter (4.7.9.1) shall be used on a panel prepared in accordance with 4.7.3 to test the dynamic COF (4.7.9.2). The dynamic COF shall be measured, in accordance with the rotating-arm meter manufacturer's instructions, three times dry, and then three wet using substitute ocean water in accordance with ASTM D1141. The average of the three measurements, dry and wet, shall be recorded. The average dynamic COF shall be in accordance with the requirements of 3.4.9.

4.7.9.1 Dynamic COF test apparatus. The dynamic COF testing device shall be Vision Point Systems Inc., µ-Deck Rotating Arm COF Meter, or equivalent system as approved by the qualifying activity.

4.7.9.2 Dynamic COF test procedure. The test shall be conducted on one test panel. The COF test panel shall be in the as-applied state (i.e., not subjected to conditioning nor wear cycles) and shall be exposed to this test procedure under the following two conditions in the sequence shown below:

a. Dry – COF test shall be performed on each test panel.

b. Wet – After completion of the dry condition test, the test panels shall be uniformly wetted with substitute seawater in accordance with ASTM D1141 from a spray bottle until the substitute seawater runs off the side of the test panel surface. The dynamic COF test shall be performed on each test panel immediately after uniform wetting with substitute seawater.

The serial number of the dynamic COF meter used in this procedure shall be listed in the qualification report. The dynamic COF meter shall be used per the manufacturer's commercial product data sheet and the following test procedure:

(1) Unlock the ball holder and insert the contact ball into the holder ensuring that the surface that shall be in contact with the surface to be measured has not been used for a previous measurement. Lock the ball into the ball holder.

(2) Place the COF meter on the test panel ensuring that the ball will remain in contact with the surface of the test panel throughout the full 360-degree rotation of the contact ball.

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(3) Measure and record the surface temperature (Ts) of the nonskid test panel in degrees Fahrenheit. The surface temperature of each COF reading shall be listed in the qualification report.

(4) Measure the uncorrected COF (COFU) in accordance with the COF meter manufacturer's instructions. COFU shall be listed in the qualification report.

(5) Calculate the temperature corrected COF (COFC) per the following equation:

COFC = COFU - 0.0048 (75 - Ts)

(6) COFC shall be listed in the qualification report.

4.7.9.3 Dynamic COF meter calibration. The COF meter shall be calibrated per the manufacturer's commercial product data sheet whenever a new contact ball is inserted.

4.8 Qualitative testing.

4.8.1 Heat resistance (all types, all classes, all grades, all compositions, all applications). Three test panels shall be prepared in accordance with 4.5.2 and coated in accordance with 4.5.2.1 or 4.5.2.2. The test panels shall be placed in an oven on a suitable rack or holder so that no parts of the test panels are in direct contact with the bottom or sides of the oven and subject to the required temperature for the coating's respective class as shown in table VII. The test panels shall be exposed to that temperature for 24 hours. At the end of 24 hours, the test panels shall be allowed to cool to standard laboratory conditions (see 4.5.1). The test results shall meet the requirements of 3.5.1.

4.8.2 Salt spray resistance (application A, all types, all classes [excluding class 4], all grades, all compositions). Three test panels shall be prepared in accordance with 4.5.2 and coated in accordance with 4.5.2.1 or 4.5.2.2. The test panels shall be placed in ASTM B117 salt spray for 96 hours. The test panels shall be allowed to air dry for 24 hours at standard laboratory conditions (see 4.5.1). The test panels shall be inspected in accordance with 3.5.2. The test panels shall be placed in an oven and exposed to the respective class temperature shown in table VII for 8 hours. The test panels shall be allowed to cool to ambient conditions (see 4.5.1) and assessed in accordance with 3.4.2. The combination of salt spray and heat exposure shall constitute one cycle. The test panels shall be exposed to a total of 5 cycles. Assessments in accordance with 3.5.2 shall be performed after 1 cycle and after 5 cycles. The test results shall meet the requirements of 3.5.2.

4.8.3 Storage stability.

4.8.3.1 Full container (all types, all classes, all grades, all compositions, all applications).

4.8.3.1.1 Type I (all classes, all grades, all compositions, all applications). Two unopened, sealed containers of type I coating shall be maintained at 120 °F (49 °C) and at a relative humidity of 80 percent undisturbed for 30 days. The container shall be opened, and the coating assessed in accordance with FED-STD-141, method 3011.3. The test results shall meet the requirements of 3.5.3.1.1.

4.8.3.1.2 Type II (all classes, all grades, all compositions, all applications). Two sealed packages of type II containers shall be maintained at 120 °F (49 °C) and at relative humidity of 80 percent undisturbed for 30 days. The container shall be opened, and the coating assessed for solidification, hard dry caking, or tough gummy residue in accordance with FED-STD-141, method 3011.3. The test results shall meet the requirements of 3.5.3.1.2.

4.8.3.2 Partially full container (type I, composition 1, all classes, all grades, all applications). One container of type I, composition 1 coating shall be opened and examined in accordance with 3.5.3.2. The lid shall be firmly replaced, and the container resealed with masking tape. Another container of the type I, composition 1 coating shall be opened and examined in accordance with 3.5.3.2. The coating shall be removed from the container until approximately ⅔ to ½ of the original volume of the coating remains in the container. The lid shall be firmly replaced, and the container resealed with masking tape. Both containers shall be tested in accordance with FED-STD-141, method 3022.2 for 48 hours. After 48 hours, both containers shall be removed and the coating examined for undesirable qualities in accordance with FED-STD-141, method 3022.2. The test results shall meet the requirements of 3.5.3.2.

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4.8.4 Sag resistance (type I, all classes [excluding class 4], all grades, all compositions, all applications). One test panel shall be prepared in accordance with 4.5.2 and coated in accordance with 4.5.2.1, except that the coating shall be applied at twice the maximum WFT cited on the manufacturer's ASTM F718 data sheet. The sag resistance of the coating shall be determined in accordance with ASTM D4400 by applying the coating material to two test panels prepared in accordance with 4.5.2 under standard laboratory conditions (see 4.5.1). The test results shall meet the requirements of 3.5.4.

4.8.5 Fluid resistance (all types, all classes, all grades, all compositions, all applications). Two test panels shall be prepared in accordance with 4.5.2 and coated in accordance with 4.5.2.1 or 4.5.2.2. The test results shall meet the requirements of 3.5.5.

4.8.5.1 Hydrocarbon. When prepared in accordance with 4.8.5, one coated test panel shall be submerged in JP-8 for 2 hours. The test results shall meet the requirements of 3.5.5.1.

4.8.5.2 Potable water. When prepared in accordance with 4.8.5, one coated test panel shall be submerged in potable water for 2 hours. The test results shall meet the requirements of 3.5.5.2.

4.8.6 Recoating (type II, all classes, all grades, all compositions, all applications). Three test panels shall be prepared in accordance with 4.5.2 and powder coated in accordance with 4.5.2.2. A 0.5-inch by 3-inch diagonal scribe to bare metal shall be made into each coated test panel. The coated test panel shall be solvent cleaned to SSPC-SP 1 cleanliness and recoated with any qualified type I, class 1, 2, or 3, grade A coating of the same class as the powder coating in accordance with requirements cited on the manufacturer's ASTM F718 data sheet. The recoated test panels shall be allowed to rest at standard laboratory conditions (see 4.5.1) for 24 hours. The recoated test panels shall be exposed to 100 hours of ASTM B117 salt spray for 100 hours followed by 8 hours of the heat requirement for the respective coating class shown in table VII. The test results shall meet the requirements of 3.5.6.

4.8.7 Brushing properties (type I, class 1, 2, and 3, all grades, all compositions, all applications). A test panel shall be prepared in accordance with 4.5.2. The applied coating shall be assessed for brushing properties in accordance with FED-STD-141, method 4321.3, as well as for flow and leveling properties, a smooth uniform film, seeds, runs, sags, or streaks. The test results shall meet the requirements of 3.5.7.

4.8.8 Spray properties.

4.8.8.1 Type I (all classes [excluding class 4], all grades, all compositions, all applications). A test panel shall be prepared in accordance with 4.5.2. The coating shall be spray applied in accordance with FED-STD-141, method 2131.2. The applied coating shall be assessed for spraying properties in accordance with FED-STD-141, method 4331.2, as well as for running, sagging, streaking, dusting, mottling, smoothness, and uniformity. Spraying properties and drying shall be in accordance with the requirements of 3.5.8.1.

4.8.8.2 Type II (all classes, all grades, all compositions, all applications). A test panel shall be prepared in accordance with 4.5.2. Powder shall be applied to the test panel at a minimum film thickness of 3 mils or applied at the minimum coating thickness cited in the manufacturer's commercial product data sheet for powder coatings. After heat curing, the coated test panel shall comply with 3.5.8.2.

4.8.9 Color retention (application A, all types, all classes, all grades, all compositions). A test panel shall be prepared in accordance with 4.5.2 and coated in accordance with 4.5.2.1 or 4.5.2.2. The panel shall be exposed for 300 hours to accelerated weathering using a Fluorescent UV-Condensation type apparatus in accordance with ASTM G154. The cycle shall be 4 hours of UV exposure using a UVB-313 bulb at 140 °F (60 °C) and 4 hours of condensation exposure at 104 °F (40 °C). At the end of the test, the panel shall be allowed to dry and compared to the identified color.

5. PACKAGING

5.1 Packaging requirements. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2).

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6. NOTES

(This section contains information of a general or explanatory nature that is helpful, but is not mandatory.)

6.1 Intended use. These materials are intended for use on substrates that experience elevated service temperatures, such as piping, exhaust manifolds, flight decks, etc. There are two types of coatings covered by this specification. Type I liquid paint is intended for use on heated steel surfaces, at temperatures up to 650 °C (1,200 °F). Type II powder coating is intended for use on heated steel surfaces, temperature of which will not exceed 372 °C (700 °F).

6.1.1 Welding practices. Neither type I nor type II coatings described by this specification are intended to be weld-through coatings. Type I and type II coatings should be removed to a distance of 6 inches (15 centimeters) prior to any welding procedure.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification, including any amendments.
b. Type, class, grade, composition, and application (see 1.2).
c. Requirement for identification characteristics, if other than specified (see 3.3).
d. Requirement for batch VOC certification.
e. Requirement for metals content .
f. Requirement for hazardous air pollutants content (see 3.3.5).
g. Quantity/unit of purchase required.
h. Requirement for manufacturer's commercial product data sheet for use in accordance with ASTM F718.
i. Requirement for safety data sheet in accordance with ANSI Z129.1 or FED-STD-313 (see 3.3.7).
j. Color (see 3.4.1 or SAE-AMS-STD-595 color designation nomenclature recommended).
k. Requirement for quality conformance, if other than specified (see 4.4).
l. Packaging requirements (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in the applicable qualified products list — whether or not such products have actually been so listed by that date. The attention of the supplier is called to this requirement, and manufacturers are urged to arrange to have the products they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or email CommandStandards@navy.mil, and information pertaining to qualification of products may be obtained from that activity.

6.4 Toxicity evaluation. The NMCPHC requires sufficient information to permit an HHA of the product. Upon completion of the HHA, a copy will be provided by the NMCPHC to the Government for evaluation. The HHA process is described on the NMCPHC's website, http://www.med.navy.mil/sites/nmcphc/industrial-hygiene/Pages/health-hazard-assessment.aspx.

6.5 Packaging recommendations. Suggested packaging and marking requirements are contained in table VIII.

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TABLE VIII. Packaging.

Packaging	Recommended requirements for direct Government acquisitions
Containers	(1) Type I paints should be specified to be furnished in cans capable of holding up to 1 pint (0.47 L), up to 1 quart (0.945 L), up to 1 gallon (3.78 L), up to 2 gallons (7.56 L), and up to 5 gallons (18.9L). (2) Friction plug containers should be in accordance with PPP-C-96, type V, class 2. Interior coatings should be as specified therein. Exterior coatings, including side seam stripping, should be as specified therein for plan B. Wire handles as specified therein should be provided for the 1-gallon container. Closure of the properly filled and sealed cans should be as specified in the appendix thereto. (3) All containers should comply with the requirements of the Uniform Freight Classifications (UFC), the National Motor Freight Classification (NMFC), and the applicable requirements of the Code of Federal Regulations 49 CFR 178, Department of Transportation (DOT). (4) Unit of procurement: The paints covered by this specification should be purchased by volume. The unit of procurement should be in multiples of 1 U.S. liquid gallon or 1 L at 60 °F (15.5 °C). (5) The powder coatings covered by this specification should be purchased by weight. The unit of procurement should be in multiples of weight units as customarily supplied by the powder coating manufacturer.
Commercial packaging	(1) Commercial packaging should be to ASTM D3951. (2) All containers should comply with the requirements of the UFC, NMFC, and the applicable requirements of the Code of Federal Regulations 49 CFR 178, DOT. (3) The components should be furnished in cans capable of holding 0.47 liter (1 pint), 0.945 liter (1 quart), 3.78 liters (1 gallon), and 18.9 liters (5 gallons). In addition, coatings may also be furnished in cartridges, burst packs, and aerosol cans as defined in the acquisition document. Kits should consist of multiple containers, one of which should be large enough to contain all components for mixing purposes. Not all components are required to be supplied in a single can.
Packing	Packing should be specified as follows: (1) Overseas delivery packing. Intermediate containers of like sizes of paint should be packed in accordance with ASTM D6251 in close-fitting wood boxes. Box closure and strapping should be as specified in the applicable box specification or the appendix thereto except that strapping should be flat and the finish B. (2) Domestic delivery (level B) packing. Level B packing should be as for level A, except that boxes should be domestic type or class and the strapping should be finish A or B. (3) Commercial packing. The paint, in the unit and intermediate containers should, as applicable, be packed in multiples of like sizes in accordance with UFC, NMFC, and 49 CFR 178 requirements.
Palletization	Intermediate containers should be palletized using MIL-HDBK-774 for guidance. Only one size unit or intermediate container should be placed on a pallet.

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TABLE VIII. Packaging — Continued.

Packaging	Recommended requirements for direct Government acquisitions
Intermediate containers	Intermediate containers should be close-fitting corrugated fiberboard boxes in accordance with UFC, NMFC, and 49 CFR 178 requirements. Fiberboard used in the construction of interior (unit and intermediate) and exterior containers, including interior packaging forms, should conform to ASTM D4727/D4727M; classes should be domestic fire-retardant or weather resistant fire-retardant.
Packing for acquisitions involving direct delivery to Navy ships or installations	Treated lumber and plywood. All lumber and plywood, including laminated veneer materials, used in shipping container and pallet construction, member, blocking, bracing, and reinforcing must be fire-retardant treated material in accordance with MIL-L-19140 as follows: (a) General use, weather resistant: MIL-L-19140, type II, category I. (b) General use, non-weather resistant: MIL-L-19140, type I, category I.
Safety data sheets (SDS) and Product / procedure data sheets	A copy of the SDS and company product data/procedure sheets should be attached to the shipping document for each destination (see 3.3.7).
VOC certification sheets	VOC certification sheets for each batch of coating should be provided when requested (see 6.2).
Bar codes	Marking should include bar codes.
Hazardous warnings	(a) Labels should be in accordance with 29 CFR Parts 1910, 1915, 1917, 1918, 1926, and 1928, as well as PPP-P-1892. (b) All individual containers should have the following marking: "CAUTION: This paint contains volatile solvents, with probable hazardous vapors. Use with adequate ventilation. Avoid prolonged breathing of vapors or spray mists. The solvents are highly flammable; avoid open flame and smoking." (c) Each container, shipping container, and palletized load should be marked with the appropriate hazardous symbol. (d) Unit containers should be marked: "This product is Asbestos, Lead, Chromium, Cadmium free, except for possible trace levels."
Volatile organic compound (VOC) content	"Contains (insert VOC content) grams per liter (insert VOC content in lb/gal) of volatile organic content per 40 CFR CH. 1, Part 60, Appendix A-7 (EPA), method 24.
Hazardous air pollutants (HAPS)	"Contains (insert HAPS content here in g/L and lb/gal) solids (non-volatiles) per 40 CFR 63."
Shelf-life	Each unit container, intermediate container where applicable, and shipping container should be marked as follows: "Date of first re-inspection (insert here date 2 years after date of manufacture)."

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6.6 Subject term (key word) listing.

Hazardous Air Pollutants
NESHAP
Non-lead
Temperature, high
Topcoat
VOC
Volatile organic content compliant

6.7 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensive changes.

MILITARY INTERESTS:

Custodians:
Army – CR4
Navy – SH
Air Force – 20

Review Activities:
Army – MD, MI, MR
Navy – AS, OS, YD

CIVIL AGENCY COORDINATING ACTIVITY:
GSA – FSS

Preparing Activity:
Navy – SH
(Project 8010-2020-016)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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